



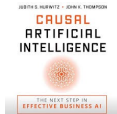
MSML610: Advanced Machine Learning

Lesson 08.1: Intro to Causal AI

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References:

- Hurwitz, Thompson: Causal Artificial Intelligence, 2024



- *Introduction and Motivation*
 - Background
 - What ML Systems Can and Cannot Tell You
- Why Causal AI Matters
- Causal AI Fundamentals
- Causal AI in Business

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Big Data and Traditional AI

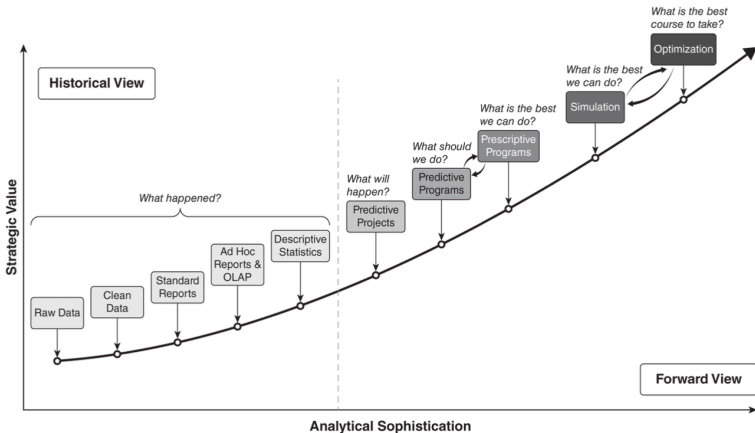
- **Focus** for past 10 years:
 - Collect and organize massive data volumes
 - Run analytics (dashboards, models, reports) on data

What Are Data Analytics?

- **Collections of data**
 - Organized datasets for analysis
 - E.g., customer purchase histories in CRM
- **Descriptive statistics**
 - Key metrics: mean, median, mode, standard deviation
 - E.g., average quarterly sales to track trends
- **Historical reports**
 - Review of *past performance*
 - E.g., monthly sales reports from last year
- **Dashboards**
 - Visuals of key metrics for quick insights
 - E.g., quarterly revenue and expense dashboard
- **Models**
 - Statistical tools to *forecast, explain*
 - E.g., churn prediction from customer behavior

Data Analytics Sophistication

Business Question	Methodology
What happened?	Descriptive statistics
What will happen?	Predictive models
What should we do?	Prescriptive programs
What's the best we can do?	Simulation + optimization



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The Illusion of Understanding

- **High accuracy $\neq$ Understanding causation**
 - 95% churn prediction accuracy won't tell you what *causes* churn
 - Accurate sales forecasts can't predict effects of *pricing changes*
- **Prediction vs. Explanation**
 - Models find patterns in historical data
 - Can't explain mechanisms or answer “what if” scenarios
- **The Real Problem**
 - Without causation, you react to past events
 - You can't proactively *change outcomes*

Association, Correlation, and Causation

- **Association**

- Variables co-occur or are statistically dependent
- E.g., “*People who buy diapers also buy beer*”
- No info on direction or mechanism

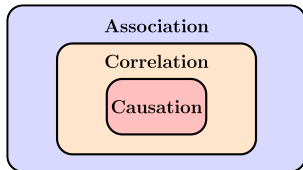
- **Correlation**

- Linear relationship between variables
- E.g., “*Ice cream sales and drowning deaths correlate*” (both seasonal)

- **Causation**

- Changing one variable *directly* changes another
- E.g., “*Asbestos causes cancer*”
- Requires causal model or experiment

- **The illusion**: high correlation *feels like* understanding
 - Real understanding means knowing what happens when you *intervene*



What ML Systems Can Tell You

- **Pattern recognition**
 - Find correlations in data
 - Identify similar items (clustering, classification)
 - Forecast based on past trends
- **Predictive accuracy on held-out test data**
 - Accurate prediction if conditions remain stable
 - Only works when training and future data match
- **Feature importance** (with caveats)
 - Find variables with strongest statistical associations
 - Doesn't distinguish correlation from causation

What ML Systems Cannot Tell You

- **Causation from observational data alone**
 - Correlation $\neq$ causation
 - Strong correlations can arise from confounding, reverse causation, or coincidence
- **Effects of interventions**
 - *“If we change X, what happens to Y?”*
 - E.g., *“If we lower prices by 10%, will revenue increase?”*
 - Observational data can't answer without causal assumptions
- **Counterfactuals**
 - *“What would have happened with a different decision?”*
 - E.g., *“Would customer churn if we offered a discount?”*
- **Fairness or causal bias**
 - A model can be statistically unbiased but causally biased
 - E.g., using race-correlated variables in decisions
- **Optimal decisions**
 - ML optimizes for accuracy, not business outcomes
 - A 90% accurate model might lead to worse decisions than an 85% accurate one, depending on the consequences of different errors

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Problem 1: Correlation is Not Causation!

- **Correlation**: statistical relationships between variables
 - Pros: Find patterns in past data to predict future
 - Cons:
 - Doesn't explain cause
 - Variables move together by coincidence or hidden factors
- **Causation**: explains how changing one variable influences another
 - Cannot be concluded from correlation alone
- **Data doesn't understand causes and effects**
 - Only humans identify variables and relationships from context
 - Without causation, you can't make intelligent decisions

Correlation is Not Causation: Examples

- Intuitively you know that association is not causation
- Humans confuse association with causation
 - “*You eat X, get stomachaches*” → infer “*X is bad for you*”
 - “*Buy stock before price spike*” → infer “*you can time markets*”
 - Leads to riskier bets
- E.g., “top-notch consulting improves businesses”
 - Or do thriving businesses hire top consultants?

Correlation is Not Causation: Examples

- E.g., in hotel industry
 - Prices are low outside tourist season
 - Prices are high when demand is high and hotel full
 - A naive prediction suggests to increase prices to sell more rooms
- E.g., online marketplace company
 - Decide how to set prices and when to have sales
 - What is the impact of lowering prices on units sold?
 - Gains from selling more need to compensate for the loss of selling cheaper
 - This is a causal question: “what would happen if we lower prices vs doing nothing”
 - Price cuts can have different impacts depending on
 - The type of business (e.g., clothing vs dog food)
 - The timing (e.g., before Christmas)

Problem 2: Decision Making

- Businesses need decisions, not just predictions
- E.g., *“What happens if a product price is reduced by 10%?”*
 - Will more customers buy?
 - If revenue decreases, what to do?
 - Why are customers leaving? Quality issue? Emerging competitor?
- **Key questions**
 - *“What happens if we change product price?”*
 - *“Low-sugar vs low-fat diet: which works?”*
 - *“If we increase credit lines, how does margin change?”*
 - *“Build a library or give tablets to boost test scores?”*

Problem 3: ML Explainability

- **Regulators require defense** of ML/AI decisions
 - E.g., hiring decisions, policy setup
- Many ML systems (e.g., neural networks) are “black boxes”
 - Humans can't understand input-to-output process
 - Can't explain decisions to shareholders
 - Features like age, race, sex introduce bias
- **Drawbacks:**
 - Fines, customer backlash, activist pressure
- **Solution:**
 - Explainable AI lets users comprehend, explain, and trust results

Problem 4: How ML Systems Make Decisions Today

- **Typical workflow**
 - Collect data → Train model → Make predictions
 - Deploy: *"If model predicts X, take action Y"*
 - Optimize for model accuracy, not decision quality
- **Critical limitation**
 - Systems optimize for prediction accuracy
 - Not for decision quality or business outcomes
 - 95% accurate fraud detector that blocks 50% legitimate transactions is a poor system

Data Science for Decisions: Examples

- **Recommendation systems** (Netflix, Amazon, TikTok)
 - Model predicts user clicks → recommends top-N items
 - Problem: Can't test interventions (e.g., diverse content for long-term satisfaction)
- **Credit scoring**
 - Model predicts default likelihood → approve/deny loans
 - Problem: Can't distinguish variables that predict vs cause default
- **Hiring pipelines**
 - Model scores candidates → HR ranks applicants
 - Problem: Training data encodes historical hiring bias
- **Fraud detection**
 - Model flags suspicious transactions → blocks automatically
 - Problem: High false positive rate without knowing how to improve

Problem 5: Feedback Loops

- When predictions change the world
- **Recommendation systems**
 - Model recommends sensational content → users click more
 - Future data: more sensational content clicks
 - Model recommends even more sensational content
 - Result: Radicalization, echo chambers, mental health issues
- **Criminal justice relapse prediction**
 - Model trained on biased arrest data (minorities over-policed)
 - Judges use scores for bail/sentencing
 - Minorities policed more → arrested more → data more biased
 - Model becomes increasingly biased over time
- **College admissions**
 - Model trained on past acceptance decisions
 - Encodes historical racism, sexism
 - Filters out biased applicants → perpetuates biases

Problem 6: Distribution Shift

- The world changes
 - **Concept drift**
 - Customer preferences change over time
 - Model trained on 2020 data becomes less accurate by 2023
 - Market conditions shift (e.g., pandemic effects)
 - **Covariate shift**
 - Input distribution changes but relationship stays the same
 - E.g., more young customers join platform
 - Model trained on older customer base performs worse
 - **Label shift**
 - Outcome distribution changes
 - E.g., fraud rate increases due to new fraud tactics
 - Model trained on old fraud rate is less effective
- **Why causal models help**
 - Encode mechanisms, not just patterns
 - Mechanisms stable across distribution shifts
 - Understand “why” → predict in new contexts

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Optimization vs. Inference vs. Decision Theory

- Three paradigms ML conflates
- **Optimization**: find best action given *known* objective
 - Assumes world behavior is known
 - E.g., minimize cost, maximize throughput, tune parameters
 - Tools: gradient descent, linear programming, evolutionary algorithms
- **Statistical inference**: estimate parameters from data
 - Learns from observations given model structure
 - E.g., estimate treatment effects, compute confidence intervals
 - Tools: regression, Bayesian estimation, hypothesis testing
- **Decision theory**: choose actions under *uncertainty*
 - E.g., “Launch product with uncertain demand?”
 - Requires reasoning about interventions and outcomes
 - Tools: utility functions, expected value of information, causal models

Where ML Falls Short

- **ML focuses on inference and optimization**
- **Real problems need decision theory**: act under uncertainty with causal knowledge

Paradigm	Question	Assumes	Output
Optimization	What maximizes $f(x)$?	Known objective	Optimal x^*
Inference	What is θ ?	Model structure	Parameter estimates
Decision theory	What should we <i>do</i> ?	Uncertainty + values	Recommended action

- **Causal AI bridges inference and decision theory**
 - Inference: what the world looks like
 - Causal models: what happens when you *act*
 - Decision theory: which action to choose

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The Cost of Ignoring Causality: Concrete Failure Modes

- When organizations ignore causality, specific and predictable failures occur
- **Simpson's Paradox**
 - Aggregate data shows opposite trend from subgroups
 - E.g., Berkeley admissions: overall bias shown, but no bias within departments
 - Causal fix: condition on right variables using DAG
- **Confounding bias**
 - Hidden variable drives both treatment and outcome
 - E.g., ice cream sales and drowning deaths both rise in summer
 - Causal fix: identify and control for confounders
- **Collider bias**
 - Conditioning on common effect creates false associations
 - E.g., hospitalized patients: unrelated diseases appear correlated (both cause hospitalization)
 - Causal fix: don't condition on colliders

The Cost of Ignoring Causality: More Failure Modes

- **Feedback loop amplification**
 - Biased predictions reinforce bias in future training data
 - E.g., predictive policing: more police to minority neighborhoods → more arrests → model learns neighborhoods are more dangerous
 - Causal fix: model feedback explicitly
- **Intervention backfire**
 - Acting on correlation causes wrong decisions
 - E.g., hospitals with more staff have higher mortality (confounded by severity)
 - Naive fix: reduce staff (catastrophic!)
 - Causal fix: distinguish correlation from causal effect before acting
- **Key takeaway**: all failures above are preventable with causal reasoning
 - Cost of ignoring causality: wrong decisions, wasted resources, real harm

Causal AI

- **Understands the why**
 - Determines cause-and-effect between variables
 - E.g., did marketing campaign increase sales?
- **Identify interventions**
 - Variables and interventions to change outcomes
 - E.g., which lifestyle changes reduce blood pressure?
- **Predict counterfactuals**
 - Hypothesize outcomes under different circumstances
 - E.g., student grades at different school?
- **Avoid bias**
 - Account for confounding variables
 - Ensure fairness beyond training data biases
- **Improve decisions**
 - Understand relationships for better choices
 - E.g., how decisions impact logistics

Causal AI vs Traditional AI

- **Current AI** uses correlation to:
 - Analyze data
 - Identify patterns
 - Make predictions
- **Model quality depends on data quality**
 - Biased or unclean data → poor model
- *“The next revolution of data science is the science of interpreting reality, not of summarizing data”* (Judea Pearl, 2021)
- **Causal inference**: Infer causation from association
 - Understand when causation and association differ
- **Normative**: Understand cause-and-effect → intervene on causes for desired effects

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From Data Science to Decision Science

- ML excels at prediction, fails at “what-if” questions
- Association works for predicting without intervention
- ML fails for decisions involving interventions—models can be useless or harmful
- **Two different paradigms**

Dimension	Data Science	Decision Science
Goal	Predict outcomes	Choose best actions
Core question	<i>“What will happen?”</i>	<i>“What should we do?”</i>
Method	Statistical models, ML	Causal models, decision theory
Output	Score, forecast, cluster	Recommended action + expected outcome
Success metric	Accuracy, AUC, RMSE	Business outcome, ROI, impact
Data requirement	Historical observations	Observations + causal structure

- **Key insight:** organizations have data science teams (prediction)
 - Business stakeholders need decision science
 - Gap between “what will happen” and “what should we do” is *causal reasoning*

Causal Questions vs. Predictive Questions

- Predictive: *what will happen?*
- Causal: *what to do? or why it happened?*

Predictive Question	Causal Question
Which patients will be readmitted?	Which intervention reduces readmission?
Which customers will churn?	What action prevents churn?
What will sales be next quarter?	How would a 10% price cut affect sales?
Which students will fail the exam?	Does tutoring improve exam scores?
Which transactions are fraudulent?	What policy changes reduce fraud?

- **Predictive**: Standard ML, observational data only
 - Useful for *monitoring* and *alerting*
- **Causal**: Causal models or experiments
 - Answer “what if” and “why”
 - Useful for *decisions* and *policy design*
- **Test**: *changing* something = causal question
 - “What happens if. . .” → causal
 - “What will happen. . .” → predictive

A Roadmap: From Prediction to Decision Intelligence

- **Level 0: Descriptive** – *“What happened?”*
 - Dashboards, reports, statistics (backward-looking)
- **Level 1: Predictive** – *“What will happen?”*
 - ML models, forecasting (assumes stable world)
- **Level 2: Causal** – *“Why?”* and *“What if we intervene?”*
 - DAGs, do-calculus (understand mechanisms)
- **Level 3: Decision** – *“What should we do?”*
 - Causal models + decision theory + business goals
- **Most organizations stuck at Level 1**
 - Causal AI enables jump to Levels 2 and 3
 - This course teaches those tools

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The Ladder of Causation

- Pearl provided a 3-level framework for understanding causality
 1. Association
 2. Intervention
 3. Counterfactuals

Level	Symbol	Activity	Typical Questions
1. Association	$\Pr(Y X)$	Observing	What is?
2. Intervention	$\Pr(Y do(X), Z)$	Intervening	What if?
3. Counterfactuals	$\Pr(Y_X x', y')$	Imagining	Why?

Rung 1: Association

- **Question:** *“How would seeing X change our belief in Y ?”*
- **Math:** $\Pr(Y|X)$
- **Activity:** Passive observation to determine if X , Y relate
 - Traditional AI/ML based on this
 - Best case: Bayesian update; worst case: overfitted model
- **Examples**
 - *“Tree has green leaves in spring”*
 - *“What does a symptom tell about disease?”*
 - *“What does a survey tell about elections?”*

Rung 2: Intervention

- **Question:** “What happens to Y if you do X ?”
- **Math:** $\Pr(Y|do(X), Z)$
- **Activity:** Understand impact of action X on Y under conditions Z
 - Requires causal model, not just observations
- **Examples**
 - “Spring makes leaves turn green”
 - “Did headache disappear from pain reliever or food?”
 - “If I take aspirin, will headache cure?”
 - “If we ban sodas, what happens?”

Rung 3: Counterfactuals

- **Question:** *“Was X that caused Y ?”*
- **Symbol:** $\Pr(Y_X|x', y')$
- **Activity:** Imagine what would happen if facts were different
 - Highest form of reasoning—requires understanding cause-and-effect
- **Examples**
 - *“What if child received adult drug dose?”*
 - *“What would jury conclude?”*
 - *“Why did marketing campaign fail?”*

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Correlation vs Causation Model

- **Correlation**: how variables relate
- **Causality**: whether one variable causes another
- Both accept inputs, transform, compute predictions
- **Correlation-based AI**: works with abundant historical data
- **Causal-based AI**: build business model first, then integrate data

Correlation-Based Model Process

- **"Data first" approach**: more data = better
- **Process**: Acquire → Clean → EDA → Feature engineering → Build → Deploy
- **Common failures**
 - Organizational issues
 - Opaque models, lack explainability
 - Spurious correlations
 - Never articulate "why ML?"

Causation-based Model Process

- **"Model first" approach**: understand business question before data
- **Process**
 - What is the intended outcome?
 - What is the proposed intervention?
 - What are confounding and effecting factors?
 - Create model diagram/graph
 - Data acquisition
 - ...

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Digital Transformation in Business

- **Digital transformation** uses digital technologies to:
 - Reduce costs
 - Improve revenues
- How?
 - **Integration of digital technology**
 - Embed digital tools (AI, cloud, IoT, automation) into business areas
 - **Cultural and organizational change**
 - Encourage innovation, agility, data-driven mindset
 - **Customer-centric approach**
 - Use digital solutions (personalized experiences, AI-driven insights) to enhance customer engagement, satisfaction
 - **Process automation and optimization**
 - Streamline operations through AI, robots, analytics
 - **Data-driven decision making**
 - Leverage big data, machine learning, real-time analytics for smarter, faster, strategic decisions

Why AI Projects Fail?

- **AI projects fail** because they approach problems only from a ML perspective
 - Data scientists:
 - Focus on model accuracy and technical metrics
 - Work in isolation from domain experts, business users, and data engineers
 - Lack of understanding of problem context leads to irrelevant solutions
- **Misalignment with business goals**
 - Models solve the wrong problem or answer irrelevant questions
- **Black-box models** fail to solve real-world problems
 - Models are difficult to interpret or explain to stakeholders
 - Lack of transparency reduces trust and adoption
- **Data issues** undermine model performance
 - Poor quality, missing, or biased data
- **Deployment and integration challenges**
 - Models not easily integrated into production systems
 - Real-time constraints or infrastructure limitations
- **Lack of feedback loops**
 - No process for monitoring model performance post-deployment

How to Make AI Project Succeed

1. Create a hybrid team

- Organizations are complex
- Single group lacks skills for difficult problems
- Hybrid team:
 - Represents all business aspects
 - Uses collaborative framework
 - Communicates with a single language (e.g., DAGs)
 - Team size varies by company and project complexity

2. Meet regularly to ensure continuity

3. Find an executive sponsor

- Understands project's goals and potential

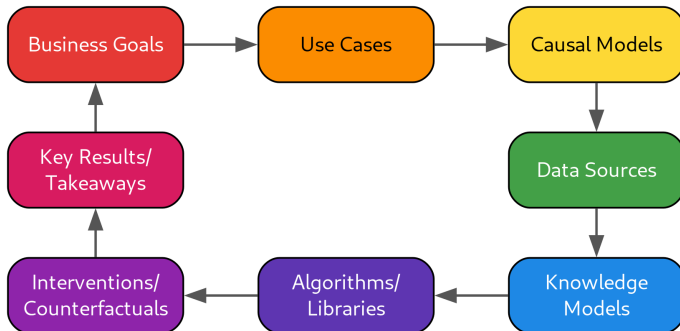
4. Initial pilot

- Small team for targeted problem
- Demonstrate AI approach merit

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AI in Business

– How do you use Causal AI in Business?



Step 1: What Are the Intended Outcomes?

- Most AI projects (even 90%) fail
 - Misalignment between business goals and project
- **Questions**
 - What process are you analyzing?
 - What happens if a course of action is taken or not?
 - Outcomes: positive, negative, unacceptable, optimal?
 - What related data can be combined and leveraged?
 - What are possible and feasible interventions?
 - Confounding factors correlated with outcomes and treatments?
 - Factors exist but cannot be accurately measured?
- ...

Step 2: What Are the Proposed Interventions?

- What are **levers** to achieve a business outcome?
- E.g., customer marketing for a product
 - Introduce a new product?
 - Does bundling multiple products
 - Improve sales?
 - Inhibit long-term sales?
 - Should advertisement focus on product quality vs other options?
 - Add more variations of the same product?
 - Discontinue the product?
 - Divest the product line?
 - ...

Step 3: What Factors Create Changes?

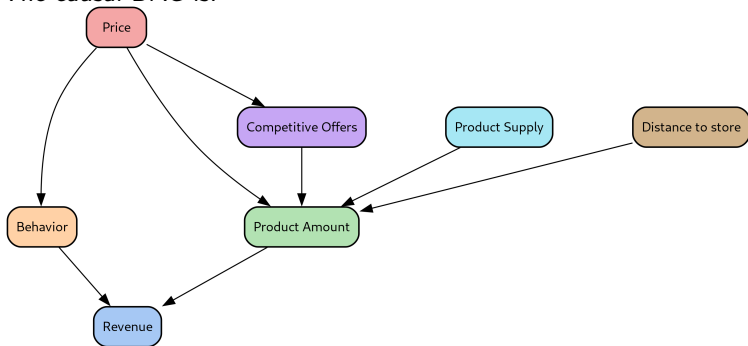
- **Identify confounders**
 - Obscure relationship between variables
 - Mute or inflate relationship
- **Direct effect**
 - Effect through intervention
- **Indirect effect**
 - Effect by environment or intervention byproduct
- **Total causal effect**
 - Effect of all factors modifying outcome

Step 4: Build Causal DAG

- **Causal models**
 - Simplify complex systems without losing key relationships
 - Focus on essential variables and interactions
- **Visual models**
 - Abstract complexity into interpretable formats
 - Highlight direction and strength of influence between variables
- **Simplicity**
 - Aid communication between technical and non-technical stakeholders
 - Reduce cognitive overload by excluding irrelevant details
 - Guide data collection by identifying impactful variables
 - Support hypothesis testing through counterfactual and intervention scenarios
- **Balance is key**
 - Too much simplicity loses insight
 - Too much complexity loses clarity

Marketing Example: Price Intervention

- Assume **price is intervention** and you want to **maximize revenue**
 - Should we increase the price and how much?
 - Should we decrease the price and how much?
 - Should price change in one-time or over time?
 - Should we develop individual pricing model for each customer?
- The causal DAG is:



- Lots of confounders!

Step 5: Data Acquisition and Integration

- Use **data collected** for correlation-based ML
- Collect **data specifically** for causal AI
 - Treat, condition, transform data
- Prepare data to match refined causal model
 - Clean, normalize, align data with model assumptions
- **Transformation data**
 - Map observed variables to DAG nodes
 - Encode categorical variables
 - Handle missing data (imputation or exclusion)
 - Normalize or scale values
- Control for bias and confounding
- **Goal:** Ensure data structure supports causal estimation

Step 6: Model Modification

- **After initial DAG** is designed
 - Refine to reduce bias and improve reliability
 - Clarify variables as confounders, mediators, or outcomes
- **Avoid common pitfalls**
 - Do not control for mediators or colliders, which can distort results
 - Control for direct and indirect confounders to prevent biased estimates
- **Test models** against technical and business objectives
- Determines necessary controls to **isolate causal effects**
 - Ensure causal assumptions align with data and domain logic
 - Improve model interpretability and predictive power

Step 7: Preparing for Deployment in Business

- Transition **from experimentation to decision-making processes**
 - Validate model against real-world business outcomes
 - Ensure stakeholders understand and trust causal logic and assumptions
- **Develop user-friendly interfaces** for business users
 - Automate data pipelines for timely updates and monitoring
- **Set up monitoring:**
 - Establish metrics for performance tracking and drift detection
 - Create feedback loops to refine and improve models post-deployment
- **Document and train:**
 - Provide clear model documentation for auditors and users
 - Train stakeholders on interpreting results and making informed decisions
- **Goal:** Deliver measurable business value

Roles in Hybrid Teams

Role	Responsibilities
Business Strategists	Align modeling with business goals Sponsor projects Communicate insights to stakeholders
Subject-Matter Experts	Provide domain expertise Identify relevant variables Validate DAGs
Data Experts	Source and clean data Map data to model variables Handle missing values
Data Scientists	Construct and validate DAGs Apply causal inference methods Simulate decisions
Software Developers	Build tools and interfaces Create data pipelines
IT Professionals	Provide infrastructure and governance Ensure model execution Integrate with enterprise systems
Project Managers	Coordinate collaboration and timelines Manage documentation Ensure alignment with strategic goals

Executing for a Hybrid Team Project

- **Establish a collaborative approach**
 - Align strategic goals with technical efforts
 - Emphasize early stakeholder engagement and shared ownership
- **Define team goals**
 - Use SMART objectives aligned with business strategy
- **Target a project**
 - Choose a bounded, feasible, high-value use case
 - Prioritize early wins for trust and momentum
- **Define the hypothesis**
 - Build a preliminary DAG with experts' input
- **Incremental model development**
 - Build the model in small stages
 - Iterate with regular feedback

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The Importance of Explainability

- **Understanding AI-based models**
 - How do ML models make decisions?
 - Can they be trusted?
 - Are they biased?
- Foster trust, transparency, and user confidence in AI systems
- Management faces
 - Loss of trust, regulation violations, fines, lawsuits
 - E.g., false negative in medical screening for cancer
- **Humans in the loop**
 - Ensure interpretation and context are considered

Techniques for Interpretability

- **Local Interpretable Model-agnostic Explanations** (LIME)
 - Focus on single prediction
 - Approximate model locally with interpretable model
 - Perturb input data, observe prediction changes
- **Partial Dependence Plots** (PDP)
 - Show marginal effect of features on predicted outcome
 - Vary one feature, keep others constant
- **Individual Conditional Expectation** (ICE)
 - Show feature-prediction relationship for individual instances
- **SHapley Additive exPlanations** (SHAP)
 - Quantify feature contribution to specific prediction

Causal AI in Interpretable AI

- **Understand causes, effects, and solutions**
 - DAGs present variables, relationships, strengths
 - Counterfactuals predict action outcomes
 - Remove confounders to prevent skewed estimates
 - Humans understand output (experts and non-experts)
 - Hybrid teams enhance context, reduce blind spots

The Future of Causal AI

- **Causal AI:**
 - Moves from academia to commercial use
 - Departs from purely data-driven systems
 - Reflects human thinking and decision-making
- **Convergence:**
 - Causal AI
 - Traditional AI
 - Deep learning
 - Generative techniques